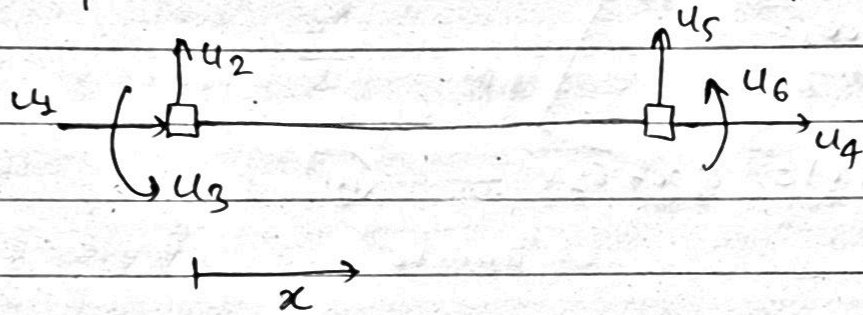


Homework: 1

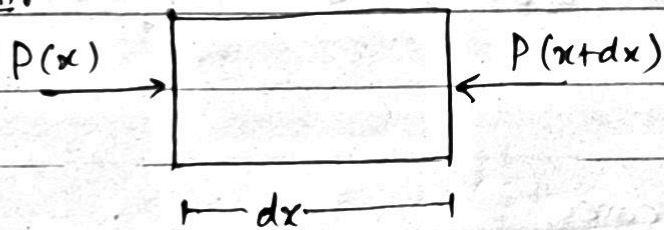
1

- ① For the two-node linear-elastic Euler-Bernoulli beam element shown below with both axial and bending degrees of freedom, derive all the components of the 6x6 element stiffness matrix. Follow the process reviewed in the class and develop the governing differential equations for axial and bending - by writing the equilibrium, compatibility, and constitutive equations and then solve for proper boundary conditions.



→ Solⁿ

- ② For axial degrees of freedom,
Assuming no distributed axial load,
Equilibrium:



$$\frac{dP(x)}{dx} = 0 \quad \text{--- ①}$$

Compatibility: strain-displacement relationship

$$\epsilon_x = \frac{du}{dx}$$

Constitutive:

for linear-elastic material,
from Hooke's law,

$$\sigma_x = E \epsilon_x \quad \Rightarrow \quad P = EA \epsilon_x = EA \frac{du}{dx} \quad \text{①}$$

from eqⁿ (I),

$$\frac{d}{dx} \left(EA \frac{du}{dx} \right) = 0$$

$$\Rightarrow EA \frac{d^2u}{dx^2} = 0 \quad \text{--- (II)}$$

Integrating twice,

$$u(x) = C_1 x + C_2 \quad \text{--- (III)}$$

Boundary Conditions

$$\text{At } x=0, \quad u(0) = u_1 \Rightarrow C_2 = u_1$$

$$\text{At } x=L, \quad u(L) = u_2 \Rightarrow C_1 L + u_1 = u_2 \Rightarrow C_1 = \frac{u_2 - u_1}{L}$$

Thus eqⁿ (III) becomes,

$$u(x) = \left(1 - \frac{x}{L}\right) u_1 + \left(\frac{x}{L}\right) u_2 \quad \text{--- (IV)}$$

from eqⁿ (I),

$$P(x) = EA \frac{du}{dx} = \frac{EA}{L} (u_2 - u_1)$$

At $x=0$, the external force must balance the internal force pointing away from the boundary,

$$F_{1x} = -P(0) = -\frac{EA}{L} (u_2 - u_1) = \frac{EA}{L} u_1 - \frac{EA}{L} u_2$$

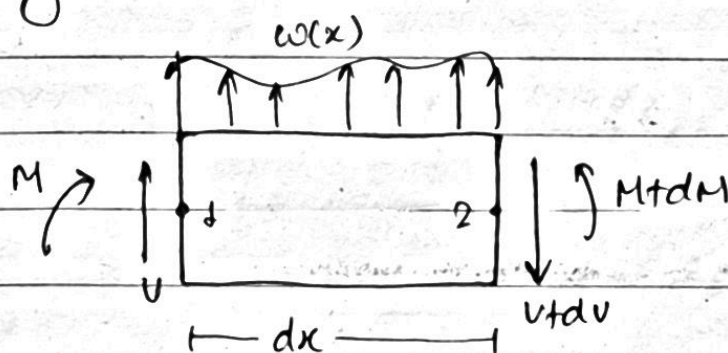
At $x=L$, the external force acts in the direction of internal tension,

$$\therefore F_{ax} = P(L) = -\frac{EA}{L} u_2 + \frac{EA}{L} u_4$$

In matrix form,

$$[K]_{axial} = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{--- (1a)}$$

(b) For bending DOFs,



Equilibrium:

$$\sum F_y = 0 \Rightarrow V - (V+dv) + w(x) dx = 0$$

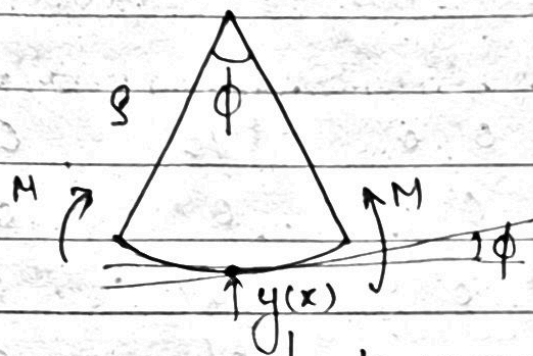
$$\Rightarrow \frac{dV}{dx} = w(x) \quad \text{--- (V)}$$

$$\sum M_2 = 0 \Rightarrow V dx + M - (M+dm) + w(x) dx \left(\frac{dx}{2}\right) = 0$$

$$\Rightarrow \frac{dM}{dx} = V \quad \text{--- (VI)}$$

Constitutive:

$$\text{Curvature } (k) = \frac{1}{\rho} = \frac{M}{EI} \quad \text{--- (VII)}$$



↳ transverse displacement function y

Compatibility:

for small angle/slope,

$$\tan \phi \approx \phi \approx \frac{dy}{dx}$$

$$\Rightarrow \phi = \frac{dy}{dx}$$

$$\text{rate of change of slope} = k = \frac{d^2y/dx^2}{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}$$

$$\approx \frac{d^2y}{dx^2} = \frac{d\phi}{dx} \quad \text{--- (VIII)}$$

from (VII) and (VIII),

$$\frac{d^2y}{dx^2} = \frac{M}{EI} \quad \Rightarrow \quad M = EI \frac{d^2y}{dx^2} \quad \text{--- (IX)}$$

And, from (v), (vi), (ix),

$$\frac{d^2 M}{dx^2} = w(x) = \frac{d^2}{dx^2} \left(EI \frac{d^2 y}{dx^2} \right)$$

$$\frac{dM}{dx} = V = \frac{d}{dx} \left(EI \frac{d^2 y}{dx^2} \right) \quad \text{--- (x)}$$

for constant EI , $V = EI \frac{d^3 y}{dx^3}$ and if only nodal forces and

moments exist, i.e. $w(x) = 0 \Rightarrow \frac{d^4 y}{dx^4} = 0$ --- (xi)

Integrating eqⁿ (xi) four times,

$$y(x) = a_4 x^3 + a_2 x^2 + a_3 x + a_4 \quad \text{--- (xii)}$$

$$\Rightarrow \phi(x) = 3a_4 x^2 + 2a_2 x + a_3 \quad \text{--- (xiii)}$$

Boundary Conditions,

$$\text{At } x=0, y(0) = u_2 \Rightarrow a_4 = u_2 \quad \text{--- (xiv)}$$

$$\text{At } x=0, \phi(0) = u_3 \Rightarrow a_3 = u_3 \quad \text{--- (xv)}$$

$$\text{At } x=L, y(L) = u_5 \Rightarrow a_4 L^3 + a_2 L^2 + a_3 L + a_4 = u_5 \quad \text{--- (xvi)}$$

$$\text{At } x=L, \phi(L) = u_6 \Rightarrow 3a_4 L^2 + 2a_2 L + a_3 = u_6 \quad \text{--- (xvii)}$$

In matrix form,

$$\{d\} = \begin{Bmatrix} u_2 \\ u_3 \\ u_5 \\ u_6 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ l^3 & l^2 & l & 1 \\ 3l^2 & 2l & 1 & 0 \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{Bmatrix} \quad \text{--- (XVIII)}$$

$$y = [N_1 \ N_2 \ N_3 \ N_4] \{d\} = [N] \{d\}$$

where, $[N]$ = Shape functions or interpolation functions known as Hermite cubic interpolation functions

Using eqⁿ (XII), (XVIII),

$$N_1 = \frac{1}{l^3} (2x^3 - 3x^2l + l^3)$$

$$N_2 = \frac{1}{l^3} (x^3l - 2x^2l^2 + xl^3)$$

$$N_3 = \frac{1}{l^3} (-2x^3 + 3x^2l)$$

$$N_4 = \frac{1}{l^3} (x^3l - x^2l^2)$$

Using eqⁿ (IX) and (X),

$$f_{xy} = V = EI \frac{d^3(y(x))}{dx^3} = \frac{EI}{l^3} (12u_2 + 6l u_3 - 12u_5 + 6l u_6) \quad \text{--- (2)}$$

$$m_1 = -m = -EI \frac{d^2 y(0)}{dx^2} = \frac{EI}{L^3} (6Lu_2 + 4L^2 u_3 - 6Lu_5 + 2L^2 u_6) \quad (3)$$

$$f_{1y} = -V = -EI \frac{d^3 y(L)}{dx^3} = \frac{EI}{L^3} (-12u_2 - 6Lu_3 + 12u_5 - 6Lu_6) \quad (4)$$

$$m_2 = m = EI \frac{d^2 y(L)}{dx^2} = \frac{EI}{L^3} (6Lu_2 + 2L^2 u_3 - 6Lu_5 + 4L^2 u_6) \quad (5)$$

Using (1a), 2, 3, 4, 5,

$$\begin{Bmatrix} f_{1x} \\ f_{1y} \\ m_1 \\ f_{2x} \\ f_{2y} \\ m_2 \end{Bmatrix} = \begin{bmatrix} AE/L & 0 & 0 & -AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{Bmatrix}$$

$$\equiv \{F\} = [K] \{q\}$$

Stiffness Matrix is symmetric because,

→ Work done by force A moving through displacement B equals the work done by force B moving through displacement A

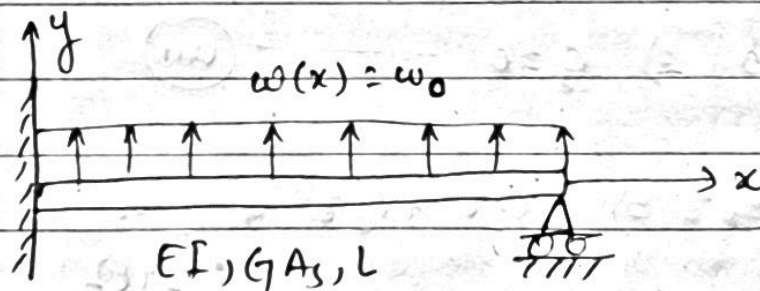
→ Proof by Betti's Reciprocal Theorem

$$F_1 \cdot u_{21} = F_2 \cdot u_{12} \quad , \quad K_{ij} = K_{ji}$$

→ In inelastic systems, the stiffness matrix is generally not symmetric because the work done is not same.

- ② Develop and solve analytically the governing differential equation for the following beam considering three cases: (a) Euler-Bernoulli beam theory, (b) pure shear beam theory, and (c) Timoshenko beam theory. For each case, derive equations for the full displacement/deformation field including vertical deformation $y(x)$, slope $\theta(x)$, and shear deformation $\gamma(x)$, and internal forces including $V(x)$ and $M(x)$.

Observe that the solution derived for the Timoshenko beam will be equivalent to the Euler-Bernoulli beam assuming $GA_s \rightarrow \infty$, and equivalent to the pure shear beam assuming $EI \rightarrow \infty$.



→ Solⁿ. (Ignoring Axial Deformation)

(a) Euler-Bernoulli Beam Theory
from problem #1,
for constant EI ,

$$EI \frac{d^4 y}{dx^4} = w(x) \quad \text{--- (I)}$$

$$M = EI \frac{d^2 y}{dx^2} \quad \text{--- (II)}$$

$$V = EI \frac{d^3 y}{dx^3} \quad \text{--- (III)}$$

$$\theta = \frac{dy}{dx} \quad \text{--- (IV)}$$

Integrating eqⁿ (I) four times,

$$y(x) = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \frac{w_0}{24EI} x^4 \quad \text{--- (V)}$$

$$\alpha(x) = \frac{dy}{dx} = C_1 + 2C_2 x + 3C_3 x^2 + \frac{w_0}{6EI} x^3 \quad \text{--- (vi)}$$

Boundary Conditions,

$$\text{At } x=0, y(0)=0 \Rightarrow C_0=0 \quad \text{--- (vii)}$$

$$\text{At } x=0, \alpha(0)=0 \Rightarrow C_1=0 \quad \text{--- (viii)}$$

$$\text{At } x=L, y(L)=0 \Rightarrow C_2 L^2 + C_3 L^3 + \frac{w_0}{24EI} L^4 = 0 \quad \text{--- (ix)}$$

$$\text{At } x=L, M(L)=0 \Rightarrow EI \left(2C_2 + 6C_3 L + \frac{w_0}{2EI} L^2 \right) = 0$$

$$\Rightarrow 2C_2 + 6C_3 L + \frac{w_0}{2EI} L^2 = 0 \quad \text{--- (x)}$$

from (ix) and (x),

$$\left(-3C_3 L - \frac{w_0}{4EI} L^2 \right) L^2 + C_3 L^3 + \frac{w_0}{24EI} L^4 = 0$$

$$\Rightarrow C_3 L = -\frac{w_0 L^2}{24EI} + 3C_3 L + \frac{w_0}{4EI} L^2$$

$$\Rightarrow 2C_3 L = \frac{w_0 L^2}{EI} \left(\frac{1}{24} - \frac{1}{4} \right) \Rightarrow C_3 = -\frac{5}{48} \frac{w_0 L}{EI} \quad \text{--- (xi)}$$

$$\begin{aligned}
 \therefore C_2 &= -3L \left(\frac{-5}{48} \frac{\omega_0 L}{EI} \right) - \frac{\omega_0}{4EI} L^2 \\
 &= \frac{15}{48} \frac{\omega_0 L^2}{EI} - \frac{\omega_0 L^2}{4EI} \\
 &= \frac{1}{16} \frac{\omega_0 L^2}{EI} \quad \text{--- (xii)}
 \end{aligned}$$

Thus,

$$y(x) = \frac{1}{16} \frac{\omega_0 L^2}{EI} x^2 - \frac{5}{48} \frac{\omega_0 L}{EI} x^3 + \frac{\omega_0}{24EI} x^4 \quad \text{--- (1a)}$$

$$\alpha(x) = \frac{1}{8} \frac{\omega_0 L^2}{EI} x - \frac{5}{16} \frac{\omega_0 L}{EI} x^2 + \frac{\omega_0}{6EI} x^3 \quad \text{--- (2a)}$$

$$\gamma(x) = 0 \quad \text{--- (3a)}$$

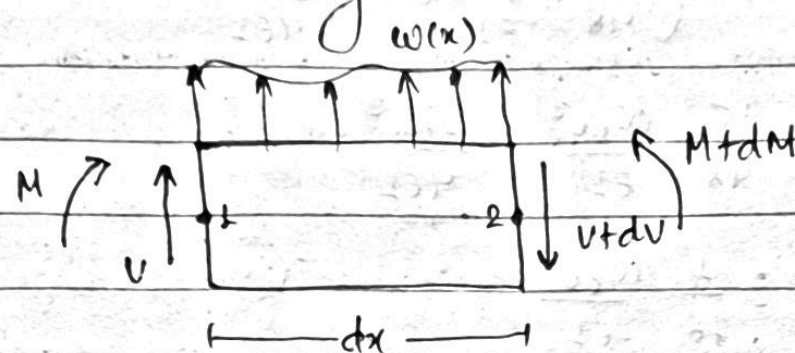
$$M(x) = EI \left(2C_2 + 6C_3 x + \frac{\omega_0}{2EI} x^2 \right)$$

$$\Rightarrow M(x) = \frac{\omega_0 L^2}{8} - \frac{5}{8} \omega_0 L x + \frac{\omega_0}{2} x^2 \quad \text{--- (4a)}$$

$$V(x) = EI \left(6C_3 + \frac{\omega_0}{EI} x \right)$$

$$\Rightarrow V(x) = \frac{-5}{8} \omega_0 L + \omega_0 x \quad \text{--- (5a)}$$

(b) Pure Shear Beam Theory



Equilibrium:

$$V + w(x)dx = V + dv$$

$$\Rightarrow \frac{dV}{dx} = -w(x) \quad \text{--- (a)}$$

$$\frac{dM}{dx} = V \quad \text{--- (a)}$$

Constitutive:

$$V = GA_s \gamma(x) \quad \text{--- (b)}$$

Compatibility:

$$\gamma(x) = \frac{dy}{dx} \quad \text{--- (c)}$$

for constant GA_s ,

$$-GA_s \frac{d^2y}{dx^2} = w(x) \quad \text{--- (d)} \quad \Rightarrow \quad \frac{d^2y}{dx^2} = -\frac{w_0}{GA_s}$$

Integrating eq (d) twice,

$$y(x) = C_4 + C_5 x - \frac{w_0 x^2}{2GA_s} \quad \text{--- (e)}$$

$$\gamma(x) = C_5 - \frac{w_0 x}{GA_s} \quad \text{--- (f)}$$

Boundary Conditions,

$$\text{At } x=0, y(0)=0 \Rightarrow C_4=0 \quad \text{--- (g)}$$

$$\text{At } x=L, y(L)=0 \Rightarrow C_5 L - \frac{\omega_0 L^2}{2GA_s} = 0 \Rightarrow C_5 = +\frac{\omega_0 L}{2GA_s} \quad \text{--- (h)}$$

Thus,

$$y(x) = +\frac{\omega_0 L}{2GA_s} x - \frac{\omega_0 x^2}{2GA_s} \quad \text{--- (1b)}$$

$$\alpha(x)=0 \quad \text{--- (2b)}$$

$$\gamma(x) = -\frac{\omega_0 L}{2GA_s} + \frac{\omega_0 x}{GA_s} \quad \text{--- (3b)}$$

$$M(x) = \int V dx \quad \text{--- (4b)}$$

$$V(x) = -\frac{\omega_0 L}{2} + \omega_0 x \quad \text{--- (5b)}$$

$$\Rightarrow M(x) = -\frac{\omega_0 L}{2} x + \frac{\omega_0 x^2}{2} + C_6$$

$$\text{At } x=L, M(L)=0 \Rightarrow -\frac{\omega_0 L^2}{2} + \frac{\omega_0 L^2}{2} + C_6 = 0 \Rightarrow C_6 = 0$$

$\therefore$ Eqⁿ (4b) becomes,

$$M(x) = -\frac{\omega_0 L}{2} x + \frac{\omega_0 x^2}{2} \quad \text{--- (4b)}$$

(c) Timoshenko Beam Theory

Equilibrium:

$$\frac{dV}{dx} = w(x) \quad \text{--- (1)}$$

$$\frac{dM}{dx} = V(x) \quad \text{--- (2)}$$

Constitutive:

$$K = \frac{1}{\beta} = \frac{M}{EI} \quad \text{--- (3)}$$

$$V(x) = GA_s \gamma(x) \quad \text{--- (4)}$$

Compatibility:

$$K(x) = \frac{d\alpha(x)}{dx} \quad \text{--- (5)}$$

$$\frac{dy(x)}{dx} = \alpha(x) - \gamma(x) \quad \text{--- (6)}$$

$\hookrightarrow$ due to Shear force
 $\rightarrow$ due to bending

$$\Rightarrow \gamma(x) = \alpha(x) - \frac{dy(x)}{dx}$$

from (1), (2), (3) and (5),
for constant EI,

$$EI \frac{d^3 \alpha(x)}{dx^3} = w(x) \quad \text{--- (7)}$$

And, (2), (4), (6) gives,

$$\frac{dy(x)}{dx} = \alpha(x) - \frac{V(x)}{GA_s} = \alpha(x) - \frac{EI}{GA_s} \frac{d^2 \alpha(x)}{dx^2} \quad \text{--- (8)}$$

Integrating eqⁿ (7) three times,

$$\alpha(x) = a_0 + a_1 x + a_2 x^2 + \frac{w_0}{6EI} x^3 \quad \text{--- (9)}$$

$$\Rightarrow y(x) = a_0 + a_1 x + a_2 x^2 + \frac{w_0}{6EI} x^3 - \frac{EI}{GA_s} \left(2a_2 + \frac{w_0}{EI} x \right)$$

$$\Rightarrow y(x) = \left(a_0 - \frac{2EI a_2}{GA_s} \right) x + \left(a_1 - \frac{w_0}{GA_s} \right) \frac{x^2}{2} + a_2 \frac{x^3}{3} + \frac{w_0}{6EI} \left(\frac{x^4}{4} \right) + a_3 \quad \text{--- (10)}$$

Boundary Conditions;

$$\text{At } x=0, y(0)=0 \Rightarrow a_3=0 \quad \text{--- (11)}$$

$$\text{At } x=0, \alpha(0)=0 \Rightarrow a_0=0 \quad \text{--- (12)}$$

$$\text{At } x=L, y(L)=0 \Rightarrow -\frac{2EI}{GA_s} a_2 L + \left(a_1 - \frac{w_0}{GA_s} \right) \frac{L^2}{2} + a_2 \frac{L^3}{3} + \frac{w_0}{6EI} \frac{L^4}{4} = 0 \quad \text{--- (13)}$$

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$$\text{At } x=L, M(L)=0 \Rightarrow \frac{EI d^2 \alpha(L)}{dx^2} = 0$$

$$\Rightarrow a_1 + 2a_2 L + \frac{\omega_0}{6EI} 3L^2 = 0$$

$$\Rightarrow a_1 = -2a_2 L - \frac{\omega_0 L^2}{2EI} \quad (14)$$

Solving (13) and (14),

$$a_2 = \frac{\frac{\omega_0 L^2}{2GA_s} + \frac{5}{24} \frac{\omega_0 L^4}{EI}}{-\frac{2EIL}{GA_s} - \frac{2L^3}{3}} \quad (15)$$

$$a_1 = -2L \left(\frac{\frac{\omega_0 L^2}{2GA_s} + \frac{5}{24} \frac{\omega_0 L^4}{EI}}{-\frac{2EIL}{GA_s} - \frac{2L^3}{3}} \right) + \frac{\omega_0 L^2}{2EI} \quad (16)$$

Thus,

$$y(x) = \frac{-2EI}{GA_s} a_2 x + \frac{a_2 x^2}{2} - \frac{\omega_0}{GA_s} \frac{x^2}{2} + \frac{a_2 x^3}{3} + \frac{\omega_0}{6EI} \frac{x^4}{4} \quad (17)$$

$$\alpha(x) = a_1 x + a_2 x^2 + \frac{\omega_0}{6EI} x^3 \quad (18)$$

$$V(x) = \frac{2EI}{GA_s} a_2 + \frac{\omega_0}{GA_s} x \quad (19)$$

$$M(x) = EI a_1 + 2EI a_2 x + \frac{w_0}{2} x^2 \quad \text{--- (4c)}$$

$$V(x) = 2EI a_2 + w_0 x \quad \text{--- (5c)}$$

This derived solution (1c), (2c), (3c), (4c), (5c) yields,

when $GA_s \rightarrow \infty$ i.e. Euler-Bernoulli Beam:

$$a_2 = -\frac{5}{16} \frac{w_0 l}{EI}, \quad a_1 = \frac{1}{8} \frac{w_0 l^2}{EI}$$

$$\therefore y(x) = 0 + \frac{1}{16} \frac{w_0 l^2}{EI} x^2 - 0 - \frac{5}{48} \frac{w_0 l}{EI} x^3 + \frac{w_0}{24EI} x^4$$

$$\alpha(x) = \frac{1}{8} \frac{w_0 l^2}{EI} x - \frac{5}{16} \frac{w_0 l}{EI} x^2 + \frac{w_0}{6EI} x^3$$

$$\gamma(x) = 0$$

$$M(x) = \frac{w_0 l^2}{8} - \frac{5}{8} w_0 l x + \frac{w_0}{2} x^2$$

$$V(x) = -\frac{5}{8} w_0 l + w_0 x$$

which are same as equations (1a), (2a), (3a), (4a) and (5a) respectively.

When $EI \rightarrow \infty$ i.e. pure shear beam,

$$a_2 = 0, a_4 = 0$$

$$2EI a_3 = -\frac{w_0 l}{2}$$

$$EI a_5 = 0$$

$$\therefore y(x) = \frac{w_0 l}{2GA_s} x - \frac{w_0 x^2}{2GA_s}$$

$$\alpha(x) = 0$$

$$\chi(x) = -\frac{w_0 l}{2GA_s} + \frac{w_0 x}{GA_s}$$

$$M(x) = -\frac{w_0 l}{2} x + \frac{w_0 x^2}{2}$$

$$V(x) = -\frac{w_0 l}{2} + w_0 x$$

which are same as equations (1b), (2b), (3b), (4b) and (5b) respectively.

Problem # 3

Consider the following properties for the beam shown in Problem #2:

Steel material: $E = 29000$ ksi, $\nu = 0.3$, $G = \frac{E}{2(1+\nu)}$

Beam section: W18x175, $h = 20.04$ in, $I = 3450$ in⁴, $A_s = 17.8$ in²

Load: $w_0 = 2$ kips/in

Consider three values for length $L = [2, 5, 20]$ ft, plot and compare the vertical deformation $y(x)$, internal shear $V(x)$, and internal moment $M(x)$ for the three cases. Study the differences as a function of h/L and $\phi = \frac{12EI}{GA_s L^2}$ and try to understand the effects of shear stiffness on the beam response. State your observations and conclusions.

Python Script

Link: https://github.com/nirajkyadav/Homeworks/blob/main/HW1/Q3_plot.py

Observations:

When the beam is long and slender, ϕ and h/L are small and bending controls the response. When the beam is short and deep, both increases and shear becomes important.

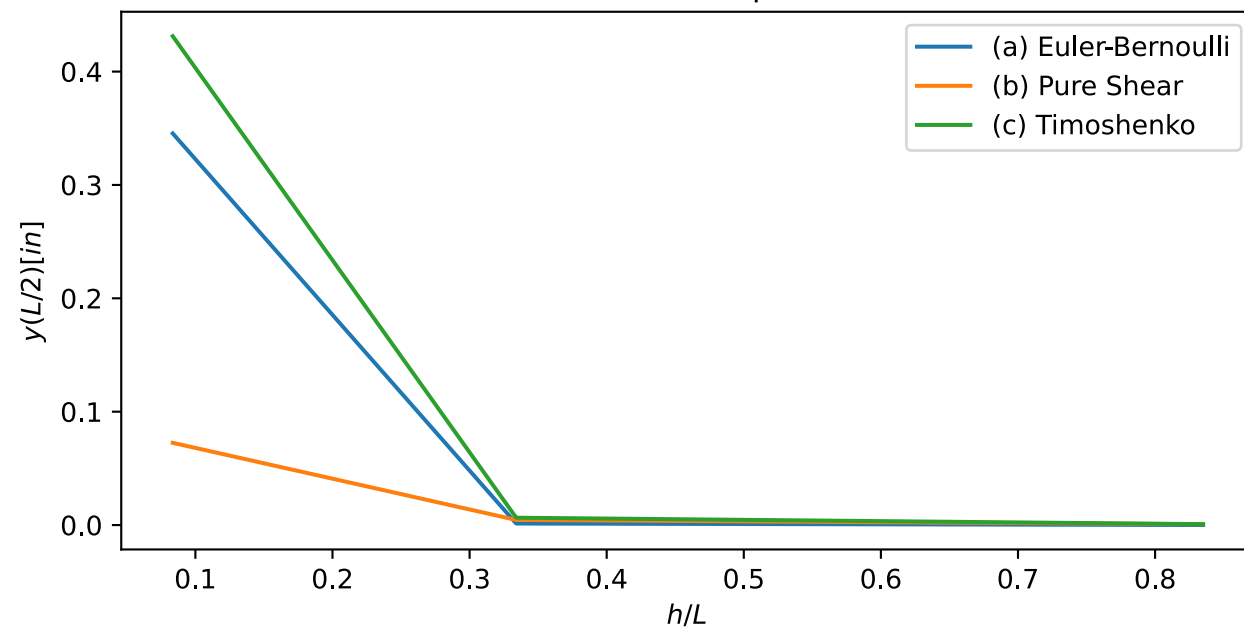
Timoshenko always gives the largest value of deflection because it carries both bending and shear deformation. Euler-Bernoulli sits just below it since it captures bending only. For the slender case the two curves are very close and shear contributes a small amount. In contrast, for short or deep beams the contribution of shear in deflection is significant. As h/L and ϕ increase, they almost overlap each other as increased depth reduces the deflection. The pure-shear model gives very little deflection but a much larger moment, so on its own it cannot represent a real beam. For internal forces, Euler-Bernoulli and Timoshenko produce the same V and M curves. That is expected as internal shear and moment come from equilibrium, not from the chosen beam theory. Their magnitudes drop as h/L rises.

Both the ϕ -plots and h/L -plots show similar patterns.

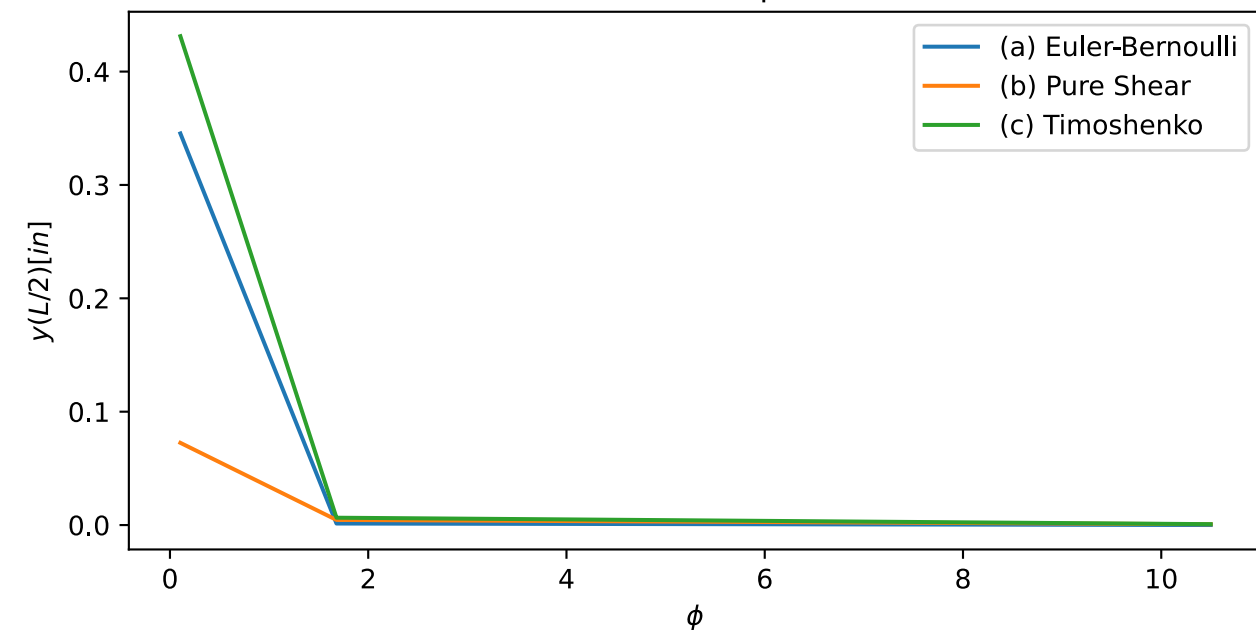
Conclusions:

When $\phi \ll 1$, shear is negligible, and if ϕ is around 1 or larger, shear should be included. Euler-Bernoulli is accurate for slender beams but underestimates deflection in deep beams because it ignores shear. Timoshenko is the safe general choice, as it collapses to Euler-Bernoulli for slender beams and correctly adds the extra shear deflection for deep ones. Pure shear alone is never adequate for a normal beam.

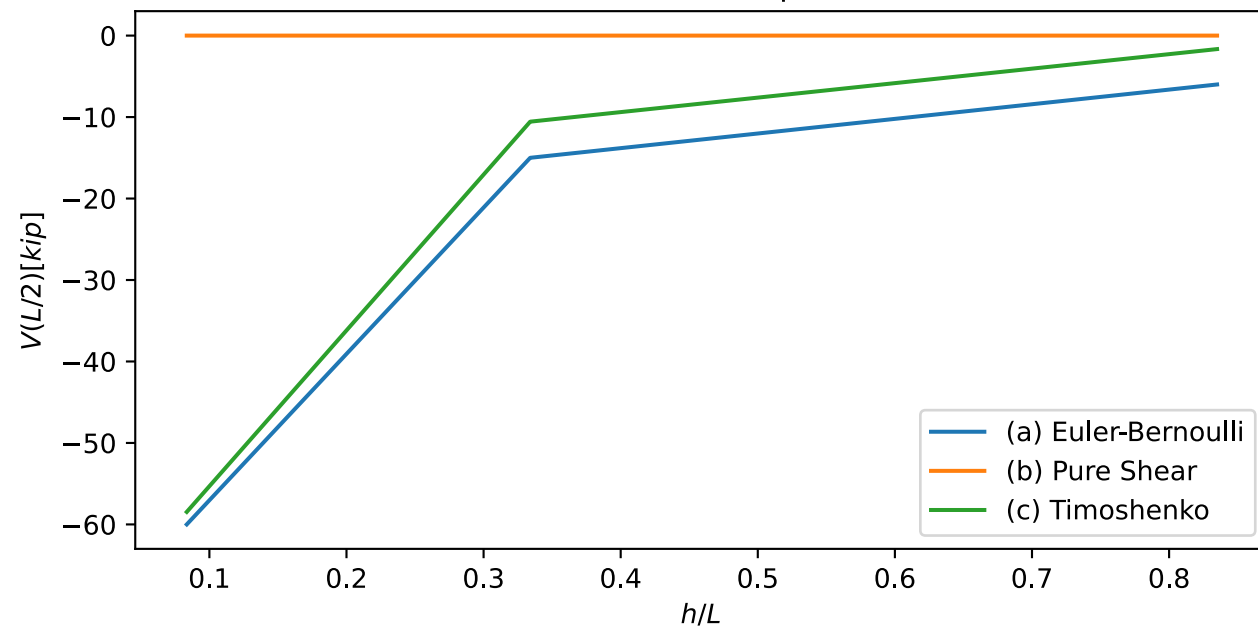
Deflection Comparison



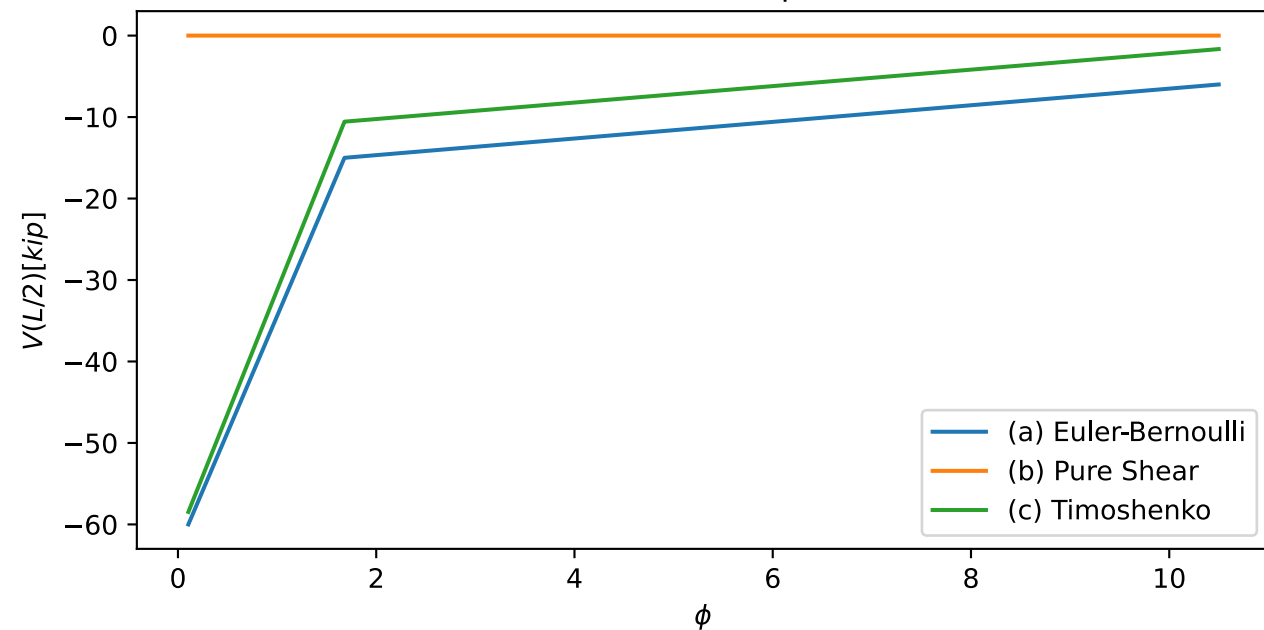
Deflection Comparison



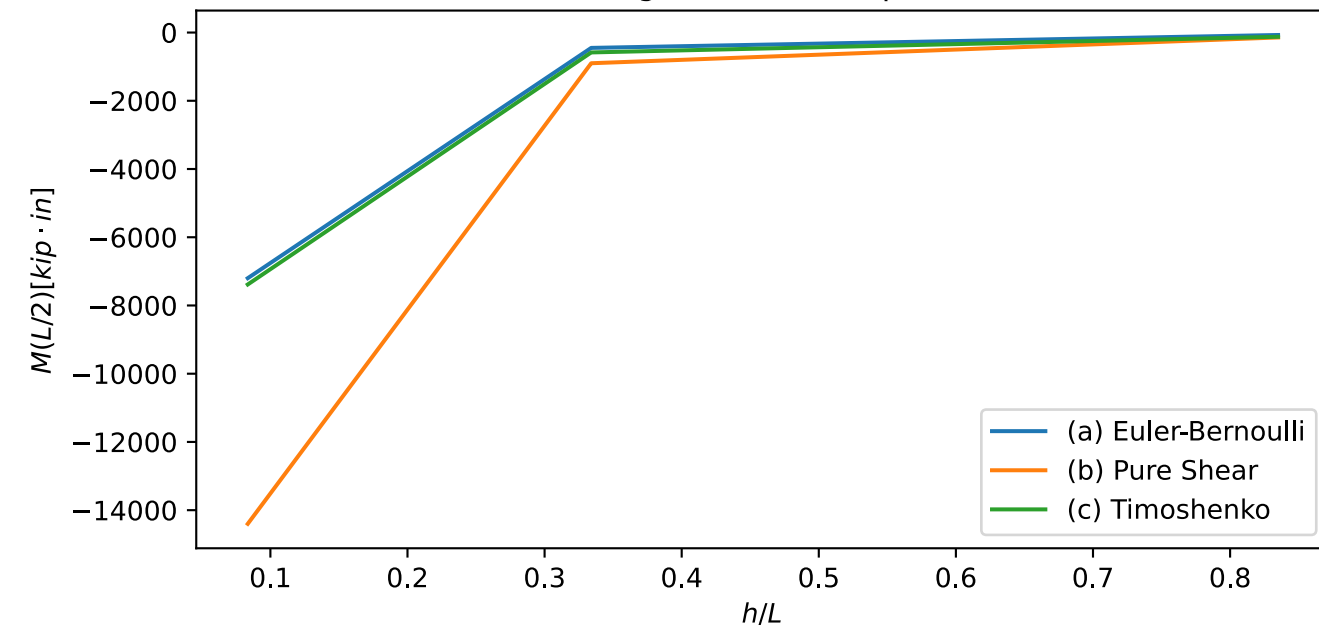
Shear Force Comparison



Shear Force Comparison



Bending Moment Comparison



Bending Moment Comparison

